Annular Collenchyma: Duchaigne (1955) reported another type called Annular collenchyma in petiole of Nerium. The lumen is more or less circular in shape.

Sclerenchyma (Gk. Sclerous- hard: enchyma-an infusion)

The sclerenchyma is dead cell and lacks protoplasm. The cells are long or short, narrow thick walled and lignified secondary walls. The cell walls of these cells are uniformly and strongly thickened. sclerenchymatous cells are of two types:

1. Sclereids
2. Fibres

Sclereids (Stone Cells)

Sclereids are dead cells, usually these are isodiametric but some are elongated too. The cell wall is very thick due to lignification. Lumen is very much reduced. The pits may simple or branched. Sclereids are mechanical in function. They give hard texture to the seed coats, endosperms etc., Sclereids are classified into the following types.

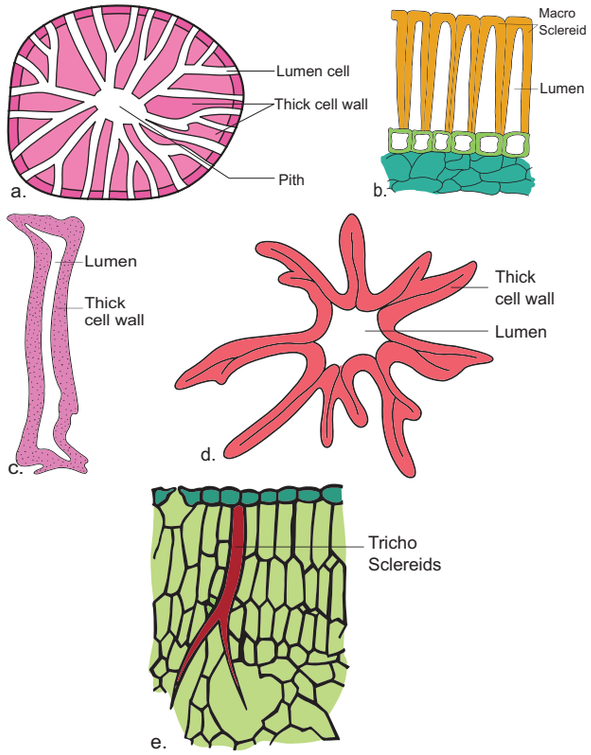


Figure 9.7: Types of sclereids
a) Brachysclereids, b) Macrosclereids, c) Osteosclereids, d) Astrosclereids, e) Trichosclereids

Types of Sclereids

1. Brachysclereids or Stone cells:
Isodiametric sclereids, with hard cell wall. It is found in bark, pith cortex, hard endosperm and fleshy portion of some fruits. Example: - Pulp of *Pyrus*.

2. Macrosclereids:
Elongated and rod shaped cells, found in the outer seed coat of leguminous plants. Example: *Crotalaria and Pisum sativum*.

3. Osteosclereids (Bone cells):
Rod shaped with dilated ends. They occur in leaves and seed coats. Example: seed coat of *Pisum and Hakea*

4. Astrosclereids:
Star cells with lobes or arms diverging from a central body. They occur in petioles and leaves. Example: *Tea, Nymphae and Trochodendron*.

5. Trichosclereids:
Hair like thin walled sclereids. Numerous small angular crystals are embedded in the wall of these sclereids, present in stems and leaves of hydrophytes. Example: *Nymphaea leaf* and Aerial roots of *Monstera*.

Sclerenchyma Found in Some Fruits



Figure 9.8: a) Pear fruit, b) Strawberry, c) Guava

Fibres

Fibres are very much elongated sclerenchyma cells with pointed tips. Fibres are dead cells and have lignified walls with narrow lumen. They have simple pits. They provide mechanical strength and protect them from the strong

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